

 **FACULTY OF ENGINEERING AND MATERIALS SCIENCE**
MECHATRONICS ENGINEERING PROGRAM

MCTR903, Winter Term 10-11

Lecture 0 – Thursday, September 23, 2010

Advanced Mechatronics Engineering

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Outline

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Course Description

Advanced Mechatronics Engineering course is designed to provide you a clear and comprehensive treatment of the various components of a mechatronics system following a multidisciplinary approach. The main topics covered in this course are:

- Measurement systems;
- Actuation systems;
- System modeling;
- Mechatronic system control;
- Interfacing systems;
- Communication systems and
- Different in-depth case studies for mechatronic systems.

Course Description

Team-based projects are integrated as an essential part of the course. These projects help you to:

- get hands-on experience in designing and implementing mechatronic systems (**Hard skills**)
- improve your competencies in communication skills, teamwork, and project management (**Soft skills**)

Course Outline

- Lecture 1: Introductory Concepts
- Lecture 2: System Modeling
- Lecture 3: System Modeling-II
- Lecture 4: Project Proposal Presentation
- Lecture 5: Signals, Systems, and Control
- Lecture 6: Signals, Systems, and Control-II
- Lecture 7: Microcontrollers
- Lecture 8: Real-time Interfacing
- Lecture 9: Communication Systems
- Lecture 10: Mechatronic Systems Case Studies
- Lecture 11: Project Presentation

Course Policy

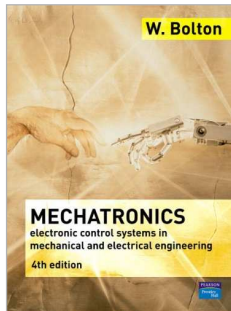
Evaluation Method		Weight	
Best 4 assignments out of 5		5%	
Best quiz out of 2		5%	
Mid-term		20%	
Project		25%	
	Project Proposal	4%	
	Proposal Presentation	4%	
	Project Final Report	5%	
	Demo	12%	
Final Exam		45%	
Total		100%	

Course Resources

- **Course Website**

TBD

- **Textbook**



W. Bolton. *Mechatronics: A Multidisciplinary Approach*. 4th Edition, Pearson, 2008.

Course Resources

- **Course Instructor**

Dr. Alaa Khamis

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- **Course TAs**

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Questions?



The ability to ask the right question is more than half the battle of finding the answer.

Thomas Watson (1874-1956)
founder of IBM